History/Final Evaluation

Demographics

Date: 07/14/2022

Patient Name: Kyle Rosebrook **Patient ID#:** 1474511

Address: 17556 West Windslow Drive Grayslake, IL 60030

Telephone (Home): (224) 540-6533 **Cell:** (224) 540-6533

Date of Birth: 01/16/1986 **Age:** 36 **Gender:** Male

Email: kylerosebrook@gmail.com

Facility

Amen Clinics Chicago

AMEN CLINICS, INC.
A Medical Corporation
2333 Waukegan Rd. Suite 150 Bannockburn, IL 60015
Phone: (224) 804-9220 Fax: (224) 241-3154
Website: www.amenclinics.com/chicago

History Date: 07/14/2022

Patient Outcome Manager: Jennifer K

Evaluation Date: 08/02/2022

Amen Clinic Physician: Michael Easton, MD

Patient Identification

Occupation: Bartender

Spouse or domestic partner's name:

Spouse or domestic partner's occupation:

Race: White

Religion:

Number of children: 1

Marital status: Single

School:

Grade:

The information presented below was obtained during an interview with Kyle, as well as a review of intake questionnaires.

Amen Clinics

How did you hear about the Amen Clinics?
Kyle heard about Amen Clinics from his mom Phillis Rosebrook.

Referral Source/Facility

Name: Phillis Rosebrook **Specialty:**
Phone: (847) 989-3275 **Fax:** **Email:**
Address:

Chief Complaint

Chief Complaint/Patient Goals for Evaluation:
"Addiction, depression."

Presenting problem/primary symptoms:
Kyle is coming to Amen Clinics for evaluation of his brain health and to address his mental health symptoms. His primary concerns include a history of addiction, depression and ADHD.

Kyle reports in 2016 when he went to college he began smoking marijuana regularly which snow balled into opiate use and over abusing benzos. He was originally prescribed Xanax for anxiety before he left for college and his use became addictive and to the extreme while away at school. He would go through his prescription in a couple of days and have to buy it off the street. In the summer of 2020, Kyle overdosed on Ambien and Xanax, causing Hypoxia. Also, in 2021, Kyle began using heroine until he overdosed in September 2021. He has not used benzos or hard drugs since. Since 2020, Kyle has noticed a substantial decrease in his memory.

Kyle reports experiencing depression since around 2008. His triggers include comparing his life to others who had completed college and had families, and his living with parents. His symptoms include isolation. Previously, he would isolate and use, now he just isolates and has low motivation to do anything. Kyle is currently sober of hard drugs since October 2022. Kyle denied thoughts, plans or intent of self harm.

Kyle reports being diagnosed with ADHD in college. If not on medication, he has extreme difficulty with focus, concentration and completion of tasks. He finds himself starting things and they fall through the cracks. He reports at times over focusing but usually an inability to focus. He experiences impulsivity and hyperactivity. Kyle reports that his organization is non-existent. He explained that his memory, both short and long term were poor before the overdoses but with worsening post incidences.

Ultimately, Kyle comes to Amen Clinics to understand the impact the substances have had and to gain support for how to improve quality of life and maintaining sobriety.

Biological Information

PRIOR ATTEMPTS TO CORRECT PROBLEMS/PRIOR PSYCHIATRIC HISTORY:
Kyle reports several treatment programs for addiction. He attended Lexing Brothers treatment program, had a stay at the Salvation Army where they supported his drug rehab and also spent a month and a half in the summer of 2020, in a residential program: Roger's Behavioral Health. Kyle has also completed two outpatient treatment programs as well.

Previous diagnoses:
Anxiety, depression, ADHD.

Medications and supplements taken at the time of scan:
None reported.

Date Started	Date Ended	Medication Name	Dosage	Times Taken Per Day	Effectiveness	Side-Effects/Problems
2010	Presently but cutting dose	Lexapro			Effective	
2010	March 2022	Buspar			Somewhat Effective	
2013	Sept 2021	Xanax			Effective	
2015	Jan 2022	Adderall			Effective	
2020	Jan 2022	Focalin			Effective	
Feb 2020	Sept 2021	Ambien			Effective	

Date Started	Date Ended	Supplement Individual or Combinations	Effectiveness	Side-Effects/Problems
2005	Present	Multivitamin	None Noticed	
Oct 2021	Present	Omega 3	None Noticed	
2005	Present	Vit C - sometimes	None Noticed	
2005	Present	Vit D - sometimes	None Noticed	

Medical History

Male

Current medical information:

Height: 5' 10"
Weight: 208 lbs
Waist: 36"

System	Past	Current	Additional Details
General		Being Overweight, Recent weight gain, Tired or worn out, Excessive Sleeping	
Neurological		Impaired ability to remember	
Respiratory			
Chest and Cardiovascular			
Head, Eye, Ear, Nose and Throat		Head injury	
Gastrointestinal			
Musculoskeletal			
Skin, Hair			
Genitourinary			
Females Only			
Males Only			
Surgical Procedures	Tonsillectomy, Adenoidectomy, Hernia repair		
Illnesses	Strep Throat, Ear Infection recently		

Past Medical Information

Reason for Hospitalization	Date / Length of Stay	Outcome
Accidental mixing of two prescriptions too close together	Sept 2021 / One day	Discharged

Prenatal and Birth Events
No pregnancy or birth complications reported.

Seizure Activity
None reported.

Exposure to Toxins
None reported.

Allergies/Drug intolerances?
Kyle reports an allergy to sulfa medications.

Head/Brain Trauma:
Kyle reports loss of consciousness and Hypoxia from a accidental Xanax and Ambien overdose in October 2021.

Tests and Labs
None reported.

Dietary/Exercise Information

Kyle's current diet is healthy with no restrictions, food allergies or food sensitivities. He reports eating healthy most of the time but sometimes eating fast food and junk box. Kyle reports a lot of walking in his current job as a bartender. He walks about five miles a shift, five days a week.

Drug and Alcohol History / Behavioral Addiction History

Concern has been expressed regarding the role of drugs in Kyle's life because of an addiction to benzos.

Kyle reports no behavioral concerns.

-Alcohol: Kyle began drinking alcohol in 2007 and drinks about once a week today. Kyle reports feeling tired and sick before getting drunk.

-Nicotine: Kyle began nicotine use in 2005 and current vapes daily.

-Marijuana: Kyle began marijuana in 2007. He used it all day, every day for many years until 2018 when his daughter was born. He will occasionally take a gummy now before bed but use is sparing.

-Cocaine: Kyle used cocaine about twenty times between 2018 and 2021.

-Opiates: Kyle used opiates intermittently in 2017.

-Benzos: Kyle began using Xanax in 2015. He was original prescribed before he left for college and his use became extreme. He would go through his prescription in a couple days and have to buy it off the street.

-Heroine: Kyle began heroine use in 2021. He overdosed in Sept 2021 and has been sober since.

Sleep Behavior

Kyle reports no problems falling asleep, staying asleep or with waking up. On average, he sleeps six and half and eight hours a night. Kyle reports heavy snoring and no sleepwalking, recurrent dreams, sleep apnea, or sleep bruxism.

Family History

Biological mother's history:
Kyle's biological mother is sixty five years old. She has no medical, learning or psychiatric problems. Her alcohol and drug use history includes occasional alcohol use. Her family history of learning, psychiatric and substance abuse include alcohol addiction.

Biological father's history:
Kyle's biological father is sixty six years old. He has no medical problems. His learning and psychiatric problems include severe depression and anxiety. His alcohol and drug use history includes past marijuana use. His family history of learning, psychiatric and substance abuse problems include severe depression and anxiety.

Siblings:
-Corey Rosebrook, 35 (Brother): Depression and anxiety.

Children:
-Kinsley Rosebrook, 3: Daughter

Psychological Information

Significant Life Events:
"Daughter born. Hospitalization for medication interaction. Car accident in 2014, was hit by a car while walking - no head injury occurred."

History of abuse:
None reported.

Social Information

Adult

Current life stressors:
"Lack of independence."

School history:
Kyle's last grade completed is his associate degree. On average, he received a 3.0. No specific learning disabilities, learning strengths or behavioral problems reported.

Employment History:
Kyle is currently employed as a bartender at Timothy O'Tools. His past jobs include mainly customer service. No work related problems reported.

Military history:
None reported.

History of legal problems:
Kyle reports one DUI.

Family structure:
Kyle lives with his mom, dad and daughter on the weekends.

Current Marital or Relationship Satisfaction:
"Single"

History of Past Marriages:
None.

Sexuality
N/A

Cultural / Ethnic Background:
"Caucasian"

Spiritual Information

Spiritual background:
"N/A"

What spiritual practices have been implemented:
"Prayer"

Unusual spiritual experience:
None reported.

Mental Status Examination

Appearance: Neat

Attitude: Cooperative

Behavior: Within normal limits

Eye Contact: Within normal limits

Speech: Within normal limits

Orientation: Kyle is oriented to person, place, time and situation.

Mood (in patient's own words): "Good, relaxed - off today and off tomorrow."

Affect: Within normal limits

Thought process: Linear

Worries/Obsessions: Denied

Delusions? Denied

Suicidal ideation? Denied

Homicidal ideation? Denied

Hallucinations: Denied

Illusions: Denied

Deja vu: No, never

Attention span: Within normal limits

Serial sevens: Done with more than one mistake 100, 93, 84, 77, 63, 54

Memory: **# of 3 remembers right away:** 3 **# of 3 remembers after 5 minutes:** 2

Judgment: Good

Abstraction:

Similarities done well

Differences done well

Relatedness: Positive

Insight: Good

Questionnaires/Checklist Results

Adult

ADULT AMEN GENERAL SYMPTOM CHECKLIST

This checklist contains a list of symptoms seen commonly in a neuropsychiatric setting. The patient and, if possible, a significant other complete it. The checklist responses suggest the following diagnoses.

According to patient:
None indicated

Other:

According to other:
Not Completed

AMEN BRAIN SYSTEM CHECKLIST

Based on his extensive brain imaging research, Dr. Amen developed the following checklist, which attempts to identify the symptoms most commonly associated with the brain systems listed below. The patient, and if possible, a significant other, complete it. The checklist responses suggest problems in the following brain systems.

According to patient:
None indicated

Other:

According to other:
Not Completed

THE AMEN CLINIC LEARNING DISABILITY ADULT SCREENING QUESTIONNAIRE

This questionnaire is a self-report form identifying possible learning disability issues in the areas of reading, writing, math, sequencing, abstraction, organization, memory, and language. The questionnaire responses suggest problems in the following areas.

According to patient:
Organization
46)My notebook/paperwork is messy or disorganized - 3
47)My room is messy - 4
48)I tend to shove everything into my backpack, desk or closet - 3
49)I have multiple piles around my room - 3
52)I often do not write down assignments or tasks and end up forgetting what to do - 3

Memory
53)I have trouble with my memory - 4
54)I remember things from long ago but not recent events - 3
55)It is hard for me to memorize things for school or work - 4
56)I know something one day but do not remember it the next day - 3

Other

According to other:
Not Completed

Other Tests Performed

Conners' Continuous Performance Task

This is a fifteen-minute computer test of attention, vigilance and impulse control. The significant findings are as follows:

Summary: None of Kyle's T-scores were atypical.

Overall, the results do not suggest that Kyle has a disorder characterized by attention deficits, such as ADHD.

Although there is no pervasive pattern of atypical T-scores, Kyle's response pattern indicate a possible issue with one or more specific dimensions of attention:
- Sustained Attention (Some Indication)

Total Brain/WebNeuro Wellness

Total Brain/WebNeuro Wellness is an objective, quantitative and standardized assessment of both symptoms and neurocognition which can help support more informed clinical decisions. The Total Brain/WebNeuro Brain Health Report identifies the patient's strengths and vulnerabilities in each of four areas: Thinking, Emotion, Self Regulation and Feeling, and gives comparisons to other healthy adults of the same age and gender. Please refer to your patient binder for a copy of your report.

PHQ-9

The PHQ-9 is a multipurpose instrument for screening, diagnosing, monitoring and measuring the severity of depression. Scores are classified as follows: 1-4 (minimal), 5-9 (mild), 10-14 (moderate), 15-19 (moderately severe), 20-27 (severe). PHQ-9 Total Score = 8

ACE QUESTIONNAIRE

The Adverse Childhood Experience (ACE) Questionnaire is a 10-item self-report measure developed to identify traumatic childhood experiences. ACE Total Score is 0/10

THE ADULT HOPE SCALE

The Adult Hope Scale is an 8 item measure of a respondent's level of hope. Scores are classified as follows: 8-16 (low), 17-39 (Slight), 40-55 (Moderate), 56-64 (High). Adult Hope Scale Total Score is 27 (Slight)

Neuropsychiatric Symptom Checklist

Please review the symptoms below and place a check in the appropriate box if you or any of your family members have had the problems listed:

Anxiety Relatives	☑ Self	☐ Mother	☑ Father	☑ Brother	☐ Sister	☐ Children	☐ Other
Panic attacks Relatives	☑ Self	☐ Mother	☑ Father	☐ Brother	☐ Sister	☐ Children	☐ Other
Phobias Relatives	☐ Self	☐ Mother	☐ Father	☐ Brother	☐ Sister	☐ Children	☐ Other
Depression Relatives	☑ Self	☑ Mother	☑ Father	☑ Brother	☐ Sister	☐ Children	☐ Other
Seasonal mood changes (SAD) Relatives	☐ Self	☐ Mother	☐ Father	☐ Brother	☐ Sister	☐ Children	☐ Other
Elevated mood Relatives	☐ Self	☐ Mother	☐ Father	☐ Brother	☐ Sister	☐ Children	☐ Other
Bipolar mood Relatives	☐ Self	☐ Mother	☐ Father	☐ Brother	☐ Sister	☐ Children	☐ Other
Mania Relatives	☐ Self	☐ Mother	☐ Father	☐ Brother	☐ Sister	☐ Children	☐ Other

Irritability Relatives	☐ Self	☐ Mother	☐ Father	☐ Brother	☐ Sister	☐ Children	☐ Other
Hot temper Relatives	☐ Self	☐ Mother	☐ Father	☐ Brother	☐ Sister	☐ Children	☐ Other
Self-mutilation Relatives	☐ Self	☐ Mother	☐ Father	☐ Brother	☐ Sister	☐ Children	☐ Other
Suicide attempts Relatives	☐ Self	☐ Mother	☐ Father	☑ Brother	☐ Sister	☐ Children	☐ Other
Psychiatric hospitalization Relatives	☐ Self	☐ Mother	☐ Father	☑ Brother	☐ Sister	☐ Children	☐ Other
Social isolation Relatives	☑ Self	☐ Mother	☐ Father	☐ Brother	☐ Sister	☐ Children	☐ Other
Hallucinations Relatives	☐ Self	☐ Mother	☐ Father	☐ Brother	☐ Sister	☐ Children	☐ Other
Schizophrenia Relatives	☐ Self	☐ Mother	☐ Father	☐ Brother	☐ Sister	☐ Children	☐ Other
Psychosis Relatives	☐ Self	☐ Mother	☐ Father	☐ Brother	☐ Sister	☐ Children	☐ Other
Paranoia Relatives	☐ Self	☐ Mother	☐ Father	☐ Brother	☐ Sister	☐ Children	☐ Other
Delusions Relatives	☐ Self	☐ Mother	☐ Father	☐ Brother	☐ Sister	☐ Children	☐ Other
Dissociative states Relatives	☐ Self	☐ Mother	☐ Father	☐ Brother	☐ Sister	☐ Children	☐ Other
Grief Relatives	☐ Self	☐ Mother	☐ Father	☐ Brother	☐ Sister	☐ Children	☐ Other
ADHD Relatives	☑ Self	☐ Mother	☑ Father	☐ Brother	☐ Sister	☐ Children	☐ Other
Concentration difficulties Relatives	☑ Self	☐ Mother	☑ Father	☐ Brother	☐ Sister	☐ Children	☐ Other
Attention difficulties Relatives	☑ Self	☐ Mother	☑ Father	☐ Brother	☐ Sister	☐ Children	☐ Other
Hyperactivity Relatives	☑ Self	☐ Mother	☐ Father	☐ Brother	☐ Sister	☐ Children	☐ Other
Intolerance of boredom Relatives	☐ Self	☐ Mother	☐ Father	☐ Brother	☐ Sister	☐ Children	☐ Other
Learning/School difficulties Relatives	☐ Self	☐ Mother	☐ Father	☐ Brother	☐ Sister	☐ Children	☐ Other
Juvenile delinquency Relatives	☐ Self	☐ Mother	☐ Father	☐ Brother	☐ Sister	☐ Children	☐ Other
Defiant behavior Relatives	☐ Self	☐ Mother	☐ Father	☐ Brother	☐ Sister	☐ Children	☐ Other
Fire setting Relatives	☐ Self	☐ Mother	☐ Father	☐ Brother	☐ Sister	☐ Children	☐ Other
Bedwetting Relatives	☐ Self	☐ Mother	☐ Father	☐ Brother	☐ Sister	☐ Children	☐ Other

Cruelty to animals Relatives	☐ Self	☐ Mother	☐ Father	☐ Brother	☐ Sister	☐ Children	☐ Other
Legal troubles Relatives	☑ Self	☐ Mother	☐ Father	☐ Brother	☐ Sister	☐ Children	☐ Other
Anger or rage problems Relatives	☐ Self	☐ Mother	☐ Father	☐ Brother	☐ Sister	☐ Children	☐ Other
Obsessions or compulsions Relatives	☐ Self	☐ Mother	☐ Father	☐ Brother	☐ Sister	☐ Children	☐ Other
Anorexia Nervosa Relatives	☐ Self	☐ Mother	☐ Father	☐ Brother	☐ Sister	☐ Children	☐ Other
Bulimia (binging/purging) Relatives	☐ Self	☐ Mother	☐ Father	☐ Brother	☐ Sister	☐ Children	☐ Other
Laxative/Diuretic abuse Relatives	☐ Self	☐ Mother	☐ Father	☐ Brother	☐ Sister	☐ Children	☐ Other
Alcohol abuse Relatives	☐ Self	☐ Mother	☐ Father	☐ Brother	☐ Sister	☐ Children	☐ Other
Drug/Substance abuse Relatives	☑ Self	☐ Mother	☐ Father	☐ Brother	☐ Sister	☐ Children	☐ Other
Head injury Relatives	☑ Self	☐ Mother	☐ Father	☐ Brother	☐ Sister	☐ Children	☐ Other
Concussion Relatives	☐ Self	☐ Mother	☐ Father	☐ Brother	☐ Sister	☐ Children	☐ Other
Tourette's Syndrome Relatives	☐ Self	☐ Mother	☐ Father	☐ Brother	☐ Sister	☐ Children	☐ Other
Amnesia Relatives	☐ Self	☐ Mother	☐ Father	☐ Brother	☐ Sister	☐ Children	☐ Other
Dementia Relatives	☐ Self	☐ Mother	☐ Father	☐ Brother	☐ Sister	☐ Children	☐ Other
Narcolepsy Relatives	☐ Self	☐ Mother	☐ Father	☐ Brother	☐ Sister	☐ Children	☐ Other
Irresistible sleep attacks Relatives	☐ Self	☐ Mother	☐ Father	☐ Brother	☐ Sister	☐ Children	☐ Other
Sleep Apnea Relatives	☐ Self	☐ Mother	☐ Father	☐ Brother	☐ Sister	☐ Children	☐ Other
Heavy snoring during sleep Relatives	☐ Self	☐ Mother	☐ Father	☐ Brother	☐ Sister	☐ Children	☐ Other
Hallucinations going to sleep Relatives	☐ Self	☐ Mother	☐ Father	☐ Brother	☐ Sister	☐ Children	☐ Other
Hallucinations when awakening Relatives	☐ Self	☐ Mother	☐ Father	☐ Brother	☐ Sister	☐ Children	☐ Other
Restless legs during sleep Relatives	☐ Self	☐ Mother	☐ Father	☐ Brother	☐ Sister	☐ Children	☐ Other
Night terrors Relatives	☐ Self	☐ Mother	☐ Father	☐ Brother	☐ Sister	☐ Children	☐ Other
Sleepwalking	☐ Self	☐ Mother	☐ Father	☐ Brother	☐ Sister	☐ Children	☐ Other

Relatives							
Sexual difficulties Relatives	☐ **Self**	☐ **Mother**	☐ **Father**	☐ **Brother**	☐ **Sister**	☐ **Children**	☐ **Other**
Sexual abuse victim Relatives	☐ **Self**	☐ **Mother**	☐ **Father**	☐ **Brother**	☐ **Sister**	☐ **Children**	☐ **Other**
Sexual abuse perpetrator Relatives	☐ **Self**	☐ **Mother**	☐ **Father**	☐ **Brother**	☐ **Sister**	☐ **Children**	☐ **Other**
Physical abuse victim Relatives	☐ **Self**	☐ **Mother**	☐ **Father**	☐ **Brother**	☐ **Sister**	☐ **Children**	☐ **Other**
Physical abuse perpetrator Relatives	☐ **Self**	☐ **Mother**	☐ **Father**	☐ **Brother**	☐ **Sister**	☐ **Children**	☐ **Other**
Mental retardation Relatives	☐ **Self**	☐ **Mother**	☐ **Father**	☐ **Brother**	☐ **Sister**	☐ **Children**	☐ **Other**
Autism Spectrum Disorder Relatives	☐ **Self**	☐ **Mother**	☐ **Father**	☐ **Brother**	☐ **Sister**	☐ **Children**	☐ **Other**
Pervasive Developmental D/O Relatives	☐ **Self**	☐ **Mother**	☐ **Father**	☐ **Brother**	☐ **Sister**	☐ **Children**	☐ **Other**
Sensitivity to light Relatives	☐ **Self**	☐ **Mother**	☐ **Father**	☐ **Brother**	☐ **Sister**	☐ **Children**	☐ **Other**
Sensitivity to odors Relatives	☐ **Self**	☐ **Mother**	☐ **Father**	☐ **Brother**	☐ **Sister**	☐ **Children**	☐ **Other**
Sensitivity to sounds Relatives	☐ **Self**	☐ **Mother**	☐ **Father**	☐ **Brother**	☐ **Sister**	☐ **Children**	☐ **Other**
Sensitivity to touch Relatives	☐ **Self**	☐ **Mother**	☐ **Father**	☐ **Brother**	☐ **Sister**	☐ **Children**	☐ **Other**

SPECT Study Parameters

As part of the evaluation, a resting and a concentration brain SPECT study were performed.

SPECT Study parameters:

The brain SPECT studies were performed in the following manner: The patient was placed in a dimly lit, quiet room. Intravenous access was obtained via small-gauge butterfly. The patient remained quiet for several minutes, with eyes open to allow their mental state to equilibrate to the environment.

For the baseline resting study, 99m Tc hexamethylpropylene amine oxime, HMPAO (Ceretec) was injected after the initial equilibration period.

For the concentration study, after the initial equilibration period, the patient started the Conners Continuous Performance Test.

A tomographic brain study was performed approximately 30-60 minutes later, using a high-resolution Picker Prism 3000 gamma camera with fan beam collimators. Data was acquired in 128 X 128 matrices. One hundred twenty images with 3 degrees separation spanning 360 degrees rotation were obtained. The data was prefiltered using a low pass filter with a high cutoff. Attenuation correction was performed using a linear method. Coronal, sagittal, and transaxial tomographs were reconstructed with a slice thickness of approximately 9 mm. The transaxial tomographs were parallel to the orbitalmeatal line. The tomographs were displayed using a standardized linear color scale. The studies were read by visual inspection in all three planes, in 3-dimensional surface brain maps (looking at the most active 45% of brain activity), and 3-dimensional active brain maps (comparing average activity with the most active 15% of brain activity).

SPECT Study Findings

Brain SPECT imaging basically shows us three things: areas of the brain that work well, areas of the brain that work too hard, and areas of the brain that do not work hard enough. With this information, together with the clinical information obtained through our extensive history-taking process, psychometric testing, and clinical evaluation, we are able to develop a more comprehensive, effective, and integrated treatment plan. SPECT scans help delineate the brain physiology underlying psychiatric problems and may

or may not fully correlate with the clinical DSM-V diagnoses, since the DSM-V is based on historical symptom clusters and not on underlying brain systems. This is part of why SPECT scans can be so helpful.

Findings:
These are good quality scans. The most significant finding is decreased tracer activity in the temporal lobes, inferior orbital prefrontal cortex, dorsal frontal cortex, parietal cortex, and the prefrontal pole accompanied by scalloping, more notably in the baseline study. Along with the decreased tracer activity in the occipital lobe and internal cerebellum, these findings may suggest a history of traumatic brain injury and an encephalopathic process. Additionally, there is severely increased tracer activity in the thalamus in both studies and mildly increased activity in the basal ganglia and insular cortex, more intense in the concentration study. There also is very mildly to mildly increased activity in the anterior cingulate gyrus and right lateral temporal lobe, which approaches a diamond-plus pattern in the concentration study. Correlation with patient history and clinical findings is recommended.

1: Decreased tracer activity in the left and right inferior orbital prefrontal cortex seen on both studies, more severe at rest, and decreased left and right lateral prefrontal cortex tracer activity seen on both studies, more severe at rest.

2: Decreased left temporal lobe tracer activity seen on both studies, more severe at rest, and decreased right temporal lobe tracer activity seen on both studies.

3: Decreased left and right parietal lobe tracer activity seen on both studies, more severe at rest.

5: Mild scalloping seen on both studies, more pronounced at rest.

6: Brain injury. A combination of findings suggests past brain injury. These findings include:
- Decreased prefrontal pole activity
- Decreased cerebellar activity
- Decreased temporal lobe activity
- Decreased parietal lobe activity
- Decreased occipital lobe activity
- Decreased dorsal prefrontal cortex activity.

7: Increased left and right basal ganglia and insular tracer activity seen with concentration.

8: Decreased internal cerebellar tracer activity seen at rest.

CONCLUSIONS:
1: Decreased tracer activity in the left and right inferior orbital prefrontal cortex seen on both studies, more severe at rest, and decreased left and right lateral prefrontal cortex tracer activity seen on both studies, more severe at rest.

When decreased activity in the inferior orbital prefrontal cortex is seen during a resting state and improves with concentration, it is often associated with depressive disorders, and may be responsive to antidepressant medication.

2: Decreased left temporal lobe tracer activity seen on both studies, more severe at rest, and decreased right temporal lobe tracer activity seen on both studies.

This abnormality may be associated with several different symptoms including mood instability, irritability, memory problems, abnormal perceptions (auditory or visual illusions, periods of deja vu), periods of anxiety or irritability with little provocation, periods of spaciness or confusion, and unexplained headaches or abdominal pain. We have found abnormalities in this part of the brain to be helped with anticonvulsant medication when clinically indicated. Decreased activity in the temporal lobes may also be associated with learning problems, especially reading comprehension difficulties and auditory processing problems. Problems in the right temporal lobe have been associated with social withdrawal, social skill struggles and depression (more inwardly directed difficulties as opposed to left sided problems). If clinically indicated, it may be helped by anticonvulsant medications. A supplement alternative might be GABA to help inhibit erratic firing. If memory problems are the primary issue, then we often prescribe acetylcholine-esterase inhibitors, Namenda or a group of supplements to enhance memory, such as gingko biloba and Phosphatidyl serine.

3: Decreased left and right parietal lobe tracer activity seen on both studies, more severe at rest.

This finding has been associated with toxic exposure, brain trauma, infection or Alzheimer's disease. The parietal lobes have also been implicated in attention issues, direction sense, performance of complex tasks and orienting oneself in space. Clinical correlation is essential.

5: Mild scalloping seen on both studies, more pronounced at rest.

This finding occurs when the brain, on the outside surface view, appears scalloped or irregular and not smooth. It is often associated with toxic exposure (such as to drugs, alcohol or environmental toxins), infection or oxygen deprivation at some point in the past. It may be seen present when overall activity is very low. We have also seen it in widespread trauma.

6: Brain injury. A combination of findings suggests past brain injury. These findings include:
- Decreased prefrontal pole activity
- Decreased cerebellar activity
- Decreased temporal lobe activity
- Decreased parietal lobe activity
- Decreased occipital lobe activity
- Decreased dorsal prefrontal cortex activity.

7: Increased left and right basal ganglia and insular tracer activity seen with concentration.

This finding may be present in normally functioning individuals. However, we also see it very frequently in people who struggle with anxiety (left sided problems are often associated with irritability, right sided problems more often associated with inwardly directed anxiety). If clinically indicated, it may be helped by antianxiety medications, such as buspirone, and deep relaxation techniques. Sometimes if the finding is focal in nature (more one side than the other) anticonvulsant medications can be helpful. When it is normal, we often see it associated with people who have high levels of motivation.

8: Decreased internal cerebellar tracer activity seen at rest.

The cerebellum is the most intensely active structure on the SPECT scan of a healthy individual. There is growing literature describing behavioral disturbances associated with abnormalities of cerebellar structure or function. This supports the belief that the cerebellum is involved in coordinating higher-order processes such as planning and impulse control and other complex behaviors because it is known to send projections to various regions of the cerebral cortex, especially the frontal and prefrontal cortices. We are currently investigating the clinical significance of this finding and to identify any associations between specific abnormalities of internal cerebellar activity and concrete behavioral disturbances

Physician Section

The above information was provided by the patient, compiled by the historian, and reviewed by me.

Your Primary Concerns:
- Depression: Baseline/ situational. Lexapro has helped.
Characterized by: Low mood, loss of interest, isolating,
feelings of negativity, ANTs (automatic negative thoughts).
- Mood instability: Situational overwhelm and defeat, avoidance, procrastination.
- Cognitive: Difficulty with sustaining attention, difficulty prioritizing, difficulty organizing, difficulty multitasking, often leaving tasks unfinished, impulsivity and hyperactivity.
- Memory difficulties: New learnings, recalling information.

Your Vision/Goals:
- Brain health assessment.
- Plan to help with mood, brain healing.

Your co-morbid medical conditions/ Biomedical ("BRIGHT MINDS") Risk Factors:
- Blood Flow: SPECT, loss of consciousness and Hypoxia from accidental Xanax and Ambien overdose in October 2021.
- Retirement/Aging: no
- Inflammation: low omega-3 intake,
- Genetics: predisposition, family history,
- Head Trauma: due to heroin, BZD OD.
- Toxins: Past polysubstance abuse,
- Mental Health: On psychiatric review of symptoms, there were no concerns for a history of psychotic disorders, mania, social anxiety, PTSD, OCD, panic disorder, eating disorders, dementia, autism.
- Immune/Infections: COVID-19 status: Pfizer x 3. ALLERGIES: sulfa medications.
- Neurohormones: unknown.
- Diabesity: Overweight
- Sleep: snoring. Variable sleep times. Non-restorative.

Kyle, here are the "Next Steps" I'd recommend you focus on first to achieve your goals and enhance your brain health:

Phase 1/ IMMEDIATE STEPS:
- TESTS.
- We ordered blood tests. Please get these done at Lab Corp or Quest. (the ones that you have not had done in the last year, and send the ones you had done, to my assistant). Please find out how much insurance will cover BEFORE getting these tests, because some insurances will only cover tests ordered by an in-network provider. At the very least, I would like CBC, CMP, vitamin D, iron panel, thyroid, DHEA, sex hormones done.
- Please get the sleep apnea test scheduled through any sleep clinic/ department that accepts your insurance. Please fill out the sleep scales sent to you (to submit to the sleep clinic to get the test scheduled).

- MEDICATION: Medication management is the standard of care. Informed consent was obtained, after addressing risks, benefits, potential side effects that are likely to influence your decision. A good website to get this information from, is the website of the US National library of medicine, National Institutes of Health. https://druginfo.nlm.nih.gov/drugportal/
- You were informed about the off-label use of medications I have recommended/ prescribed.
- If your insurance does not cover a medication, get the free GoodRx app (often, prices out of pocket are cheaper than through insurance). You can also go to https://costplusdrugs.com, which was launched in 2022 by Mark Cuban.
FIXED TIMINGS OF MEDICATIONS: SET ALARMS/ REMINDERS. Always take your night time medications and supplements at the time you start your wind-down routine (an hour before your desired bed time).
START: Pristiq 50mg every morning.
START: Gabapentin (Neurontin) 300-900mg, 1 to 3 x daily as needed for anxiety.

- SUPPLEMENTS: Please look at label information for risk and potential side-effects prior to use. For Swanson vitamins, www.swanson.com, and for BrainMD supplements, go to www.brainmd.com/chicago
- For a 15% discount from BrainMD, use the code CHICAGO15.

- BRAIN HEALTH: Brain and Body Power Max - 2 packets per day for 2 weeks. THEN 1 packet per day AND add Omega-3 power (Total fish oil needs to be a combined EPA + DHA >3000mg daily)
- Magnesium GLYCINATE. You could do chewables (deep sleep, relaxation) 300mg at night AND 300mg in the morning and afternoon as needed (from BrainMD or Swanson Vitamins).
- L-Theanine (relaxation, focus) 200mg daily (every morning). Then after 2 weeks, consider increasing to 300mg daily.
- Ashwagandha (stress, adrenals) 600-1200mg, three times a day OR JUST AT NIGHT (from Jarrow OR from Havasu nutrition).
- Brahmi bacopa 950mg every morning.
- Craving control: from brainMD (3 caps twice daily)

Phase 2 and general brain health:
- ORGANIZATION: Please work on structuring your days (ESPECIALLY unstructured time) like we discussed. Structure is KEY. SET MULTIPLE ALARMS/REMINDERS FOR: a fixed wake up time, fixed meal times, fixed exercise times, fixed times for medications/ supplements, with leaving other slots for chores, appointments, social things.

- NUTRITION: FIXED MEAL TIMES: Set alarms.
AT THE VERY LEAST: High protein, greens. Minimize carbs, and sugar and processed foods. Choose grass-fed meat.
Please ensure adequate fluid intake, at least eight 8-ounce glasses, which equals about 2 liters, or half a gallon daily. This is called the 8 X 8 rule and is very easy to remember.

- MEDITATION: Set an alarm. Fixed time (morning or winding-down before bed).
Daily meditation for as little as 15 minutes a day, can show measurable increases in grey-matter in the frontal lobes and hippocampus, and reduced grey-matter in the amygdala (which correlates with less stress).
The Headspace app also has 'SleepCasts' which are stories that help you drift into sleep. The 'Calm' app, the "10% Happier" app, and the 'breathe to relax' app are some of my favorites. See what you prefer and stick to it.

- SLEEP: Let's work on getting 7.5 hours a night. (5 sleep cycles, each cycle is 90 minutes).
7.30am: FIXED wake up time. Set an alarm AND GET OUT OF BED. If you tend to feel groggy in the morning, get some sunlight for 20 mins, have a glass of warm water, do deep breathing (30 full, slow, deep breaths) and jumping jacks or stretches.
For DINNER, minimize fast carbs/ sugars. Sweet potatoes, carrots (slow carbs) and kombucha are ok.
11pm: (1 hour before bedtime): Wind-down before bed (DIM LIGHTS, wear sunglasses indoors, scheduled worry time, scheduled to-do lists, nightly treasure hunt, meditation, boring reading, warm shower, sauna, hyperbaric oxygen, etc.)
12am: OR LATER: Only get to bed when sleepy, if possible, sleep on your NON-DOMINANT SIDE.
- When you are working late (till 3am), wake up at 7.30am the next day, and schedule a 3 hour nap that afternoon (instead of sleeping in).

- EXERCISE: FIXED exercise TIME: Set an alarm.
MODERATE intensity exercise for 30 mins, 5-7 times per week as a baseline. The simplest way, is to use a stationary bicycle or elliptical or treadmill, or even brisk walking, as if you are running late.

- SUBSTANCE USE: need to be completely avoided/ minimized.

Phase 3/ FUTURE CONSIDERATIONS:
- Please continue to work with other providers on your treatment team, as each has a role to play.

- There is a big difference between over thinking and problem solving.
- There is a big difference between catastrophizing and contingency planning.
- Success, your purpose is the journey, the process of self-discovery and self-improvement.

- Focus on the PROCESS, the outcomes will follow.

- First identify negative thoughts. THEN: Remember to: Ask yourself 'does this thought help or harm me? Does this thought save me or sabotage me? Does this thought help me achieve my stated goals or make me feel stuck? Is this the chump or the champ?' REJECT what harms you. REJECT what does is not consistent with your goals/ values. REJECT what the chump says because you know he is a liar. SAY 'SHUT UP CHUMP'.

- One of my favorite psychologists, Jordan Peterson, who is a visiting professor at Harvard, has some very insightful videos on YouTube, where he blends psychology, philosophy, dreams and life coaching seamlessly to apply to the current state of our lives and the world in general. Definitely worth listening to him (Though it may not be for everybody). He also has a best-selling book '12 rules for life, an antidote to chaos.'

- Please practice the task organization skills we discussed. There are 4 types of tasks. Divide a paper into 4 parts (quadrants), and in each part write the following:
1. Urgent AND important (these are the things you have to DO ASAP.
2. Not urgent AND important (these are the things you plan, put in your calendar or FocusMate.
3. Urgent BUT not important (for example, if Facebook reminds you it is somebodys birthday). DISCARD until you do 1 and 2.
4. Not urgent BUT not important (for example, checking your phone multiple times an hour). DISCARD.

- To be held accountable, and overcome procrastination, a website www.focusmate.com matches users for free, with other users who have a specific time and date to work on an assignment, and you hold each other accountable.

- Consider the book/ online program 'Getting Things Done' by David Allen. https://gettingthingsdone.com/

- I'm looking forward to re-connecting with you in around 4 weeks to see how you're doing on these initial steps, answer any questions that have arisen, and to work with you over the next 9-12 months to create the scaffold that will get you to that next level of brain health, functioning, and fulfillment.
All communications are handled by my assistant at drcreadoasst@amenclinic.com.
You can also call the Chicago clinic at 224 804 9220.

- It was a real pleasure meeting you today, Kyle!

Safety Plan
You denied current suicidal or homicidal or self-harm intent, ideations or plans. Safety plan: if you are becoming manic or depressed or having thoughts of hurting yourself or others, please call a mental health hotline (call 800-273-TALK (8255), or text message 741741), or call 911 or have somebody take you to the nearest ER.

Diagnosis

Current

R94.02 - Abnormal brain scan
07/18/2022 Shane Creado, MD

F90.9 - Attention-deficit hyperactivity disorder, unspecified type
07/18/2022 Shane Creado, MD

F33.1 - Major depressive disorder, recurrent, moderate
07/18/2022 Shane Creado, MD

F39 - Unspecified mood [affective] disorder
07/18/2022 Shane Creado, MD

F13.21 - Sedative, hypnotic or anxiolytic dependence, in remission
07/18/2022 Shane Creado, MD

F11.21 - Opioid dependence, in remission
07/18/2022 Shane Creado, MD

Bio-Psycho-Social-Spiritual Evaluation

Given the pattern of symptoms and scan findings, I believe there is real hope for significant improvement with a regimen to properly optimize brain function.

Biological Plan

General Bio-Medical Principles:

- Eliminate any potential toxins such as marijuana, excessive alcohol, other drugs, nicotine, caffeine, and environmental toxins.
- Treat or eliminate any potential underlying medical problem (for example: hypothyroidism, hormone imbalances, chronic infections).
- Avoid any behaviors that put your brain at risk.

Laboratory Tests: We want to rule out any underlying biomedical issues that may be causing or exacerbating your symptoms.

☑ Lab Work Recommended

In Closing

I am available by email or phone. Simple questions may be handled by a brief exchange of messages; otherwise, it is better to schedule appointments. While we do not charge for very brief issues requiring five minutes or less of my time, letter writing, form completion, record review, review of laboratory results, medication prior authorization, and other requests outside of scheduled appointments will incur a charge, depending on the amount of time needed.

I am happy to speak to and collaborate with anyone involved in your care. Please sign a release for me to send a copy of this report to any health care professional you are currently seeing, and anyone else you would like to receive one.

Consider a follow-up scan in 6-12 months to check progress. In our experience, a follow up scan can be a very powerful aid to track your progress and fine-tune your treatment.

Thank you very much for allowing me to participate in your care! I am optimistic that the recommendations we discussed will be very helpful to you. If you have any questions, please contact my assistant at drcreadoasst@amenclinic.com.

Shane Creado, M.D.

General Psychiatry, Sports Psychiatry, Sleep Medicine

Diplomate, American Board of Psychiatry and Neurology

Physician Signatures